Viable Bowel Probe to Measure Bowel Impedance

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Introduction

- Mesenteric ischemia is characterized by insufficient blood flow to the mesenteric vessels.[1] It is considered an emergency surgical case with death rates of 50-80% (Cleveland Clinic)
- It is pertinent to remove only the necessary bowel as extreme bowel resection can lead to digestive problems, infection, and tissue death
- The current method to identify bowel is to dye the GI system with Indocyanine green (ICG) dye. When exposed to fluorescent light, the green intensifies in areas with sufficient blood supply

Problem Statement

- Because of the high urgency and severity of bowel ischemia, it is critical that surgeons can gather as much information about the patient's condition in the shortest amount of time possible.

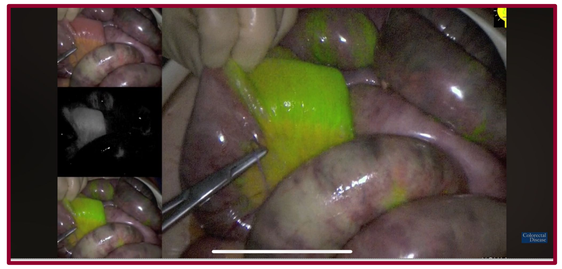


Figure 1. Usage of ICG dye in bowel ischemia surgery. Extracted from Colorectal Disease Journal[2]

Approach

- Previous studies have shown promising results for accurately identifying viable bowel through measuring its electrical properties using a probe on tissue surface[3].
- Worked out conversions of current and the tissue impedance to calculate a bowel viability based on research[3, 4]. Also determined optimal frequency to measure at.

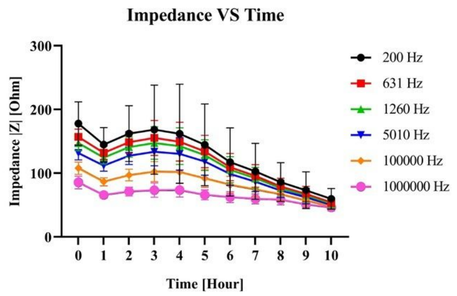


Figure 2. Measured impedance of bowel tissue over time with error bars for different frequencies from Hou J et al. 2021. It was found that with a sampling frequency of 1MHz, the error bars were the smallest. Therefore for the circuit, the current sensor chosen must ideally be able to measure at a sampling frequency of 1MHz.

Primary Focus for the Semester

- Identify an alternative mode to detect bowel viability to act as a supplement or replacement to ICG dye for physicians Proof of concept of working electrical circuit to measure impedance.
- Integrating human factors engineering to make the probe ergonomic for the healthcare professionals to use.
- Prioritize patient safety by building a device that gives the surgeon more data points to narrow the scope of the procedure they need to execute

Methods

Prototype / Design of Probe

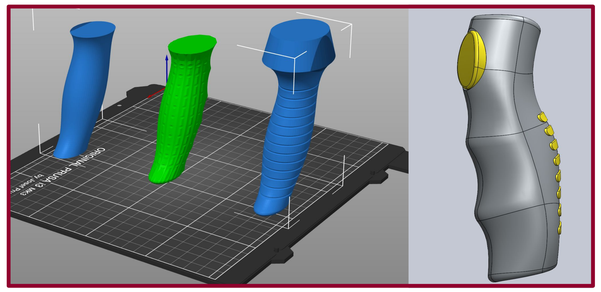


Figure 3. CAD Designs of various ergonomic handles for bowel probe. There are ridges and grooves built in based on literature reviews of comfortable handle holding. Designs made in SOLIDWORKS.The handle diameter was 41 mm and handle length was 140 mm.

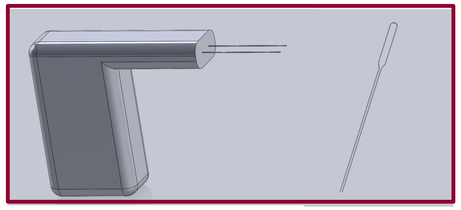


Figure 4. CAD Design for Prototype 1 of probe. Utilizes Bipolar electrodes. Intended to be handheld by healthcare personnel and battery run.

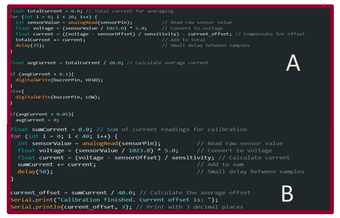
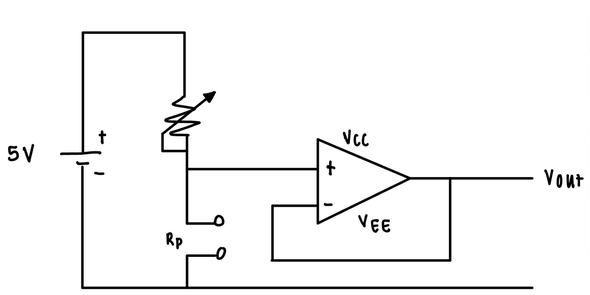


Figure 5. Code for viable bowel probe. 5A) Calibration code to account for circuit's internal resistance. 5B) Testing code that takes the data from the current sensor and converts it to impedance.

- Handheld device with electrodes that records current and calculates impedance
- Bipolar electrodes are designed to be inserted into the probe and discarded afterwards; they are single use.
- When placed over an area of dead tissue, probe will buzz and emit a visual signal to indicate to physicians bowel that can be taken out during mesenteric ischemia

Circuit Schematic Proof of Concept



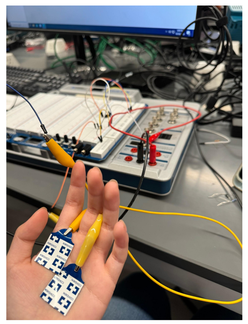


Figure 6. (a) Circuit schematic and pinouts for current sensor. The current source of 5V from the ARDUINO is in series with a potentiometer and the bipolar electrodes. This is connected to an op-amp to amplify the current coming from the electrode. The op-amp is placed to increase the resolution. (b) Proof of concept of circuit schematic in **6a** as the resistance of skin was tested. The resistance of the skin was measured to be 45 kΩ which is in range of literature values.

Results

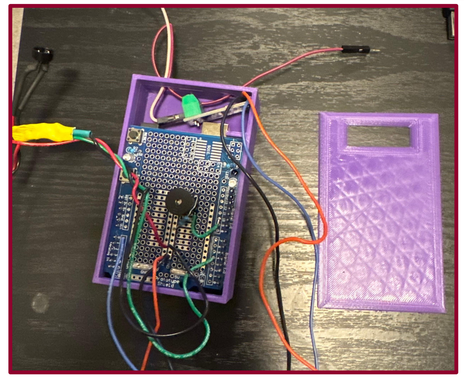


Figure 7. Current functioning prototype as of 5/5/2025.

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Starting sensor calibration...
Calibration finished. Current offset is: -0.374
Current: 0.000 A
Current: 0.000 A
Current: 0.071 A
Current: 0.000 A
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Figure 8. Snippet of Code Output

- Prototype currently outputs on average 0A. The current sensor that we had on hand was had a minimum current sensitivity threshold of 0-50A.
- The maximum current that can be sent through a tissue without affecting it is 10mA.[4]
- Current prototype has the electrodes as single use and does not address biocompatibility or reusability needs.
- Current Sensors we used: INA219, CS70331, INA333
  - All had a resolution of 50A over 1024 bits
  - Needed: 0.1 mV/A change
  - 0.4 mV/A change

Future Directions

- Test on resistors and skin resistances to ensure circuit correctly measures impedance over live tissue.
- Fully assemble CAD Body with working current sensor with the correct resolution.
- Receive physician feedback for usage

Key Takeaways

- Because electrical activity is indicative of blood flow and metabolic activity[5], it poses as the next best alternative to work in tandem with the ICG dye to characterize the bowel tissue is funcional.
- However, it is pertinent to account for the roadblocks associated with accurately measuring current. Obtaining the proper sensors to measure current within a set parameter will cause the probe to not produce any readings
- Thus, it's important that next steps include using different current sensors to test on live tissue and various viscosity of surfaces, as the bowel itself is covered in a layer of fluid that can also hinder readings.

References

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